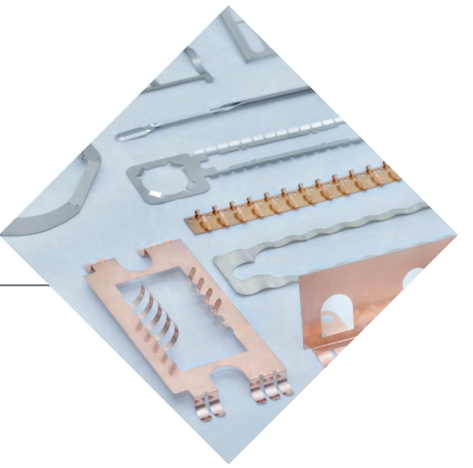
TECH|ETCH



PRECISION ETCH & FORM

TECH ETCH specializes in photo etching parts from difficult to etch specialty materials, such as Elgiloy, Titanium, Polyimide and Nitinol, which have characteristics attractive to the medical industry for implants, springs, cathodes, blades, and stents. Parts can be made from metals as thin as .0005".

STANDARD MATERIALS

- Copper
- Beryllium Copper
- Stainless Steels
- Aluminum
- Nickel Silver
- Brass
- Phosphor Bronze
- Kovar
- Iron Alloys
- Inconel

SPECIALTY MATERIALS

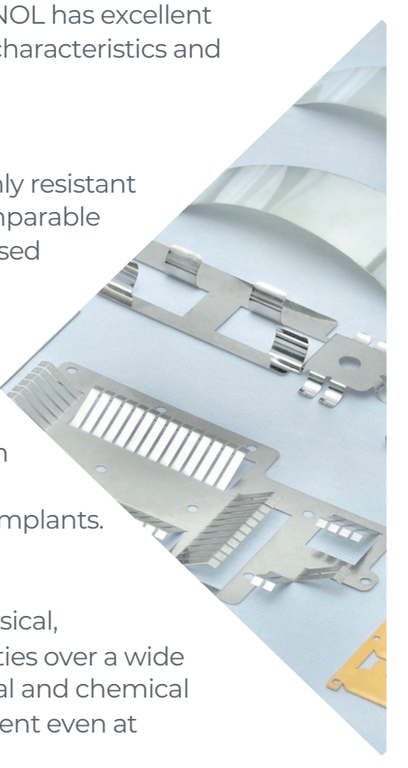
- Tungsten
- Molybdenum
- Titanium
- Elgiloy™
- Kapton® (Polyimide)
- Niobium
- Nitinol (NiTi)
- Tantalum

In addition to photo etching, Tech Etch offers a wide variety of in-house capabilities including laser cutting for thicker materials, forming, laminating, and stamping.

Formed parts are manufactured by combining photo etching, used for blanking, with inexpensive or universal tooling, used for forming.
Heat treatment is available to enhance spring qualities.

PROPERTIES

- **TUNGSTEN & MOLYBDENUM**
  Are difficult to etch refractory materials used for high-temperature, corrosion resistant applications.
- **NIOBIUM (Columbium)**
  Is a light weight refractory material with excellent high-temperature corrosion resistance. It is ductile and easily formed.
- **NITINOL (NiTi)**
  Is a shape memory alloy. Its unique characteristics allow it to return to a predetermined shape after undergoing deformation. NITINOL has excellent biocompatibility, good spring characteristics and high corrosion resistance.
- **TITANIUM**
  Is strong, light weight and highly resistant to corrosion. Its strength is comparable to 304 stainless steel and it is used for human implants.
- **ELGILOY**
  Is typically used when requirements call for a material that is highly corrosion resistant with high fatigue strength. It is used for human implants.
- **POLYIMIDE**
  Is a film that exhibits good physical, chemical and electrical properties over a wide temperature range. Its electrical and chemical resistance properties are excellent even at unusually high temperatures.



techetch.com/precision-etch-form

ISO 9001:2015
AS9100D
ISO 13485:2016
REGISTERED

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A TOTAL CAPABILITY

World Technology Leader

Tech Etch manufactures precision parts for the medical device and equipment, military/aerospace, electronic, telecommunications, and measurement and control industries. Tech Etch specializes in photo etching, forming and laminating engineered components, in addition to flexible circuits. The company has a long history of technological innovation, quality products, excellent customer service, and growth. From its beginnings as a research firm in 1964, Tech Etch has expanded its product offerings, acquired new businesses, and constructed new facilities. Tech Etch operates three manufacturing facilities and has over 700 employees. Corporate headquarters and most of the custom engineering and manufacturing are located in Plymouth, MA, just 40 miles south of Boston. Over 190,000 square feet of floor space, conference and training rooms, and state-of-the-art chemical processes and equipment make this Tech Etch's centerpiece. The other facilities are located in Fall River, Massachusetts and Litchfield, Minnesota.

Wide Range of Processes

Tech Etch performs a wide variety of processes, and this single-source capability enables it to assume total responsibility for the quality and delivery of our precision products. In-plant services include photo etching and chemical milling, artwork generation and phototooling, stamping from coil stock and forming from etched blanks, tool and die making, production heat treating, flexible circuit design and production, welding and soldering, metal finishing, plating, and laser cutting. Secondary operations such as soldering joints to seal seams, spot welding, and the application of pressure sensitive tapes and insulation materials are also available.

ISO 9001, ISO 13485, and AS9100D Certification

Tech Etch operates a Quality Management System that is registered to ISO 9001, ISO 13485, and AS9100D. This system is designed to control all business processes from Contract Review through the manufacturing processes to final inspection of product. The system seeks to identify and eliminate all nonconformities and prevent their recurrence through Corrective Action and Internal Auditing procedures. Process capability studies are available to guide you on part manufacturability.

Innovation Center

Tech Etch is leading the way to help facilitate you in quick prototyping by means of our state-of-the-art Innovation Center. The center is equipped with leading edge technology, expert engineering team, and top of the line software. Tech Etch is your one stop shop to take your idea and get it to production. For more information visit techetch.com/innovation-center.

★ ADDITIONAL PRODUCTS & CAPABILITIES ★

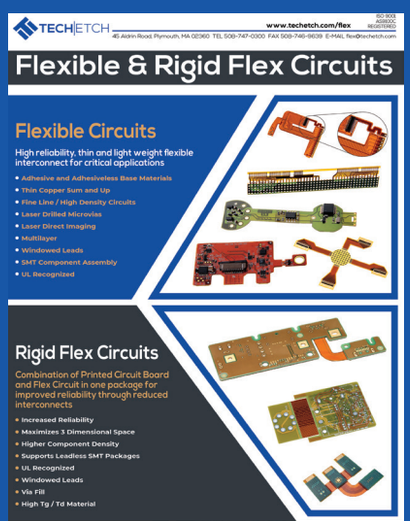
EMI/RFI SHIELDING

Tech Etch designs and manufactures a broad line of standard and custom EMI/RFI shielding products including BeCu and wire mesh gaskets, conductive elastomers, and board level shielding. Our catalog of standard products is available online or in print.



FLEXIBLE & RIGID FLEX CIRCUITS

Tech Etch has the tools and talents required to create highly reliable flex and ridged-flex circuits for a variety of uses. We also offer SMT component assembly on these products.



PRECISION ENGINEERED PARTS

Tech Etch manufactures a wide variety of lightweight metal parts as small as .0005" in thickness. We use a photo-etching process that allows us to create intricate and detailed pieces that can't be made with other processes.

